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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	"Hack into Google, No mistake"
Project / Solution Name	SOPBox: Smart Receiving Check for F&B SMEs
Name (Leader)	Lye Wei Lun
Email & Phone (Leader)	dellralllye@gmail.com & 018-960 6378
University Affiliation(s)	Tunku Abdul Rahman University of Management and Technology (TAR UMT)
Live Prototype URL	https://sopbox.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/Dellrall/SDCIdea26

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

SOPBox is a mobile-friendly web application designed to help food and beverage (F&B) SMEs check incoming deliveries. The frontend uses React, TypeScript, Vite and Tailwind CSS and communicates with a Python FastAPI REST API.

The backend manages accounts, workspace access, product catalogues, purchasing, deliveries, receiving checks, approval decisions and audit records using PostgreSQL. Workspace-scoped database queries and PostgreSQL Row Level Security are used to separate workspace data.

For photo-based extraction, the backend can send delivery-order images to a configured RunPod Serverless endpoint running vLLM with the Qwen3-VL-8B-Instruct-FP8 model. The returned data is parsed and validated before staff review and confirm the extracted lines. Deterministic application rules check product requirements, temperature limits, pack units and quantities. Failed checks require a human decision, and accepted deliveries update stock records.

The application integrates with the Telegram Bot API for configured notifications. Google sign-in and Telegram sign-in are optional configuration-dependent features, while WebAuthn supports passkey authentication. Manual entry remains available when AI extraction is unavailable. Google ADK integration, barcode scanning and A2A functionality are not implemented in the reviewed repository.

The live deployment, database configuration and external-service availability must be verified separately.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.11	Backend language	PSF License
FastAPI >=0.110	Backend REST API	MIT
React / React DOM ^18.3.1	Frontend UI	MIT
TypeScript ^5.5.0	Typed frontend code	Apache-2.0
Vite ^7.0.0	Frontend development/build	MIT
Tailwind CSS ^4.0.0	Frontend styling	MIT
PostgreSQL	Database	PostgreSQL Licence
RunPod + vLLM	AI model serving framework	Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Serverless API	Sends delivery-order photos for AI extraction	Sponsor-provided only if official credits/access were provided; otherwise paid usage-based or credits, as applicable.
Telegram Bot API	Sends delivery alerts and supports optional Telegram sign-in	Free
Google OAuth / OpenID Connect	Optional Google sign-in and account linking	Browser API; not a paid external AI API
WebAuthn browser API	Passkey registration and authentication	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Hosts and deploys the SOPBox web application and FastAPI service	Replit core
Replit PostgreSQL	Stores workspace, user, product, delivery, stock and audit data	Replit core
RunPod Serverless	Runs vLLM and the Qwen3-VL version-language model	Serverless endpoint; GPU pool documented as AMPERE_24; verify current configuration
Github	Source-code repository and version control	Repository: SDCIdea26
Telegram	Delivers notifications to configured chats	Bot API; chat IDs and token are configuration-dependent

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen3-VL-8B-Instruct-FP8	Qwen model served through vLLM on RunPod Serverless	Pre-trained inference for extracting product names, quantities and units from delivery-order images; no fine-tuning declared	Apache-2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
SOPBox illustrative delivery and product data	Created for the project demonstration	Synthetic	Team-created; no third-party dataset licence claimed	N
Product-condition demonstration images	Google Gemini image-generation tools	Synthetic	Google Generative AI Additional Terms of Service	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
ChatGPT Plus and Pro	Architecture discussions, technical troubleshooting, code review guidance, presentation wording and declaration drafting	Major
Google Gemini 3.8 Flash	Generating illustrative images of damaged and undamaged products	Minor
Claude Opus 5.5, Claude Sonnet 5.5	Engineer and implement plans into features and fixing those issues. Implement feedbacks from teammate and performing test	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-assisted outputs were reviewed by team members before use. Code and documentation were checked against the intended project design, with automated tests and manual checks used where available. Within the application, extracted document content is parsed and validated, staff confirm extracted lines, and deterministic application rules determine receiving-check outcomes rather than relying on the AI model alone. The team remains responsible for the accuracy, security, licensing and suitability of all submitted work.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core access	Development and deployment of the SOPBox prototype.
RunPod	Compute credits or GPU access	Hosting and testing the Qwen3-VL model.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

The Replit platform supports development and deployment of the application and its PostgreSQL database. RunPod provides the compute environment for image extraction. These services are replaceable in principle, but moving the deployment, database and model endpoint would require configuration and integration work. The receiving rules and approval workflow are implemented in the application code and do not depend on the AI model making business decisions. Only resources officially provided through the challenge should be identified as sponsor-provided.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

SOPBox processes account and workspace information, product catalogues, delivery details, receiving results, approval decisions and audit records. The database schema supports delivery-photo handling. The repository documents scrypt password hashing, random session tokens, an HttpOnly browser session cookie, workspace-scoped queries and PostgreSQL Row Level Security (RLS) to isolate workspace data.

When photo extraction is enabled, delivery-order images are sent to the configured RunPod endpoint. Telegram notifications may disclose delivery identifiers, supplier or staff names, failed checks and decision reasons to configured chats. Google sign-in data is processed if that optional feature is enabled. Users should avoid uploading unnecessary personal information. A complete retention and deletion schedule has not been established in the reviewed project information. Deployed security settings, provider data-handling terms, backup retention and deletion procedures must be verified.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The vision-language model may misread handwriting, quantities, units or unclear images and its output may be incomplete or invalid. SOPBox mitigates this by parsing and validating the model response, flagging unclear values instead of intentionally guessing, requiring staff to confirm extracted lines and applying deterministic product rules for temperature, quantity and pack conversion. Failed checks are held for a manager's decision and manual entry is available if extraction is unavailable or rate-limited.

The application depends on the configured RunPod endpoint for photo extraction and Telegram configuration for external notifications. Cold starts may delay extraction and provider outstage, API limits or costs may affect availability. Model output is not treated as the final safety decision. The prototype has not established measured reductions in food waste or improvements in food safety. These outcomes require real pilot and evaluation. Security and privacy controls should be tested in the deployed environment before wider use.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [x] Any use of official SDC sponsor resources has been declared in Section 5.
- [x] The prototype described here is accessible to judges through the links provided in Section 1.
- [x] All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature] 

[Date] 24/10/2026